



ADANI GROUP POWERS INDIA'S SOLAR SURGE, SHAPES GLOBAL ENERGY TRANSITION

As India cements its position as the world's renewable energy growth engine, the diversified conglomerate is not only fuelling domestic ambition but also influencing the global solar and clean energy markets

India's energy transition is no longer confined to domestic policy—it has become a global narrative. With the government committing to net-zero emissions by 2070 and a near-term target of 500 GW (gigawatt) of non-fossil fuel capacity by 2030, the country has moved into the spotlight as one of the world's fastest-growing green energy markets. India has already achieved a key milestone in its climate commitment by reaching 50 per cent of its installed power generation capacity from non-fossil sources, a full five years ahead of the 2030 target.

At the heart of this shift lies solar power. From being a marginal contributor a decade ago, solar has now emerged as the single largest component of India's renewable mix. As of March 2025, India's installed solar capacity stood at 105.65 GW, and by August 2025, cumulative capacity touched over 123 GW, accounting for nearly a quarter of India's total power generation capacity.

The pace of growth is extraordinary. Solar capacity has expanded more than 40-fold in just over a decade—from 2.8 GW in 2014 to over 100 GW in 2025. In the financial year (FY) 2024–25 alone, India added 23.83 GW of solar capacity, setting a new national record. In June 2025, the country installed 7.3 GW in a single month, a whopping 420 per cent year-on-year surge. This blistering pace makes India a global case study on how scale, policy, and innovation can align to deliver results. For policymakers in both developed and emerging economies, India is offering a model for rapid decarbonisation at scale.

A Nation of Scale, Sunshine and Strategy

India's solar story is underpinned by its natural and policy advantages. With 300–330 sunny days a year and a theoretical solar potential of over 5,000 trillion kWh (kilowatt per hour) annually, the country is endowed with abundant solar resources.

Policy support has been equally critical. Flagship schemes like PM Surya Ghar (rooftop solar adoption) and PM-KUSUM

(solarisation of agriculture pumps) have broadened adoption beyond utility-scale projects. Equally important is the push toward manufacturing self-sufficiency. Policy impetuses such as Production Linked Incentive (PLI), Approved List of Models and Manufacturers (ALMM) have helped scale solar module manufacturing from a modest 2.3 GW in 2014 to over 100 GW today. It is projected to reach 190 GW by 2027.

This shift positions India as a credible alternative in the solar supply chain—a geopolitical imperative as global economies seek to diversify sources of critical clean-tech equipment.

How Adani is Driving the Surge

At the heart of this transition are Adani portfolio companies—Adani Green Energy Limited (AGEL), renewable Independent Power Producer (IPP) and Adani New Industries Limited (ANIL), the largest integrated manufacturer of solar and wind energy equipment in India.

AGEL has emerged as India's largest renewable energy company and one of the fastest-growing clean power players globally. Adani Solar, part of ANIL is India's leading vertically integrated solar manufacturer developing a 10 GW integrated ecosystem at Mundra, Gujarat. It has an existing 4 GW cell and module manufacturing capacity, India's first and only 2 GW Ingot and Wafer manufacturing facility alongside ancillary components. As of mid-sept 2025, AGEL's operational portfolio exceeds 16 GW, enough to power over 8 million homes across 12 states. In FY 2024–25, the company added 3.3 GW of greenfield capacity, bringing its installed capacity at the end of March 2025 to about 14.2 GW. By May 2025, this figure had already risen to 14,528 MW (megawatt), underscoring its unmatched execution pace.

AGEL has committed to building 50 GW of renewable capacity by 2030. This would represent 10 per cent of India's total 2030 non-fossil fuel energy target, making the company a national anchor in meeting global climate goals.

Khavda Mega Project

- **Planned Capacity:** 30 GW
- **Area:** 538 km²
- **Power reach:** Can power over 17.4 million homes annually (equivalent to # of homes in Poland, Canada or Spain)
- **Green job:** 15,200+

SOURCE: AGEL

Khavda: The World's Largest Renewable Energy Project

AGEL's flagship project, the Khavda Renewable Energy plant in Gujarat, epitomises India's scale advantage. Spread across 538 square kilometres—an area nearly five times the size of Paris—Khavda will be the world's largest renewable energy installation, with a planned capacity of 30 GW, including 26 GW of solar.

Construction began in April 2023, and in less than a year, AGEL had already operationalised 2 GW of solar capacity at the site. By mid-2025, the cumulative operational capacity at Khavda has reached over 6 GW reinforcing AGEL's reputation for speed. Once completed, Khavda alone will supply over 87 billion units of clean and affordable power, avoiding over 63 million tonnes of CO₂ emissions annually, helping decarbonise industries, cities, and rural grids.

Khavda is more than a power plant—it is a laboratory for next-generation technologies including hybrid solar-wind setups, advanced grid balancing, and large-scale storage. It provides the blueprint for ultra large-scale renewable energy projects worldwide.

National Footprint, Global Ambition

AGEL's expansion is not confined to Gujarat. It's over 16 GW operational wind, solar and wind-solar hybrid capacity is spread across 12 states. The company is credited with developing several landmark projects in the global renewable energy (RE)